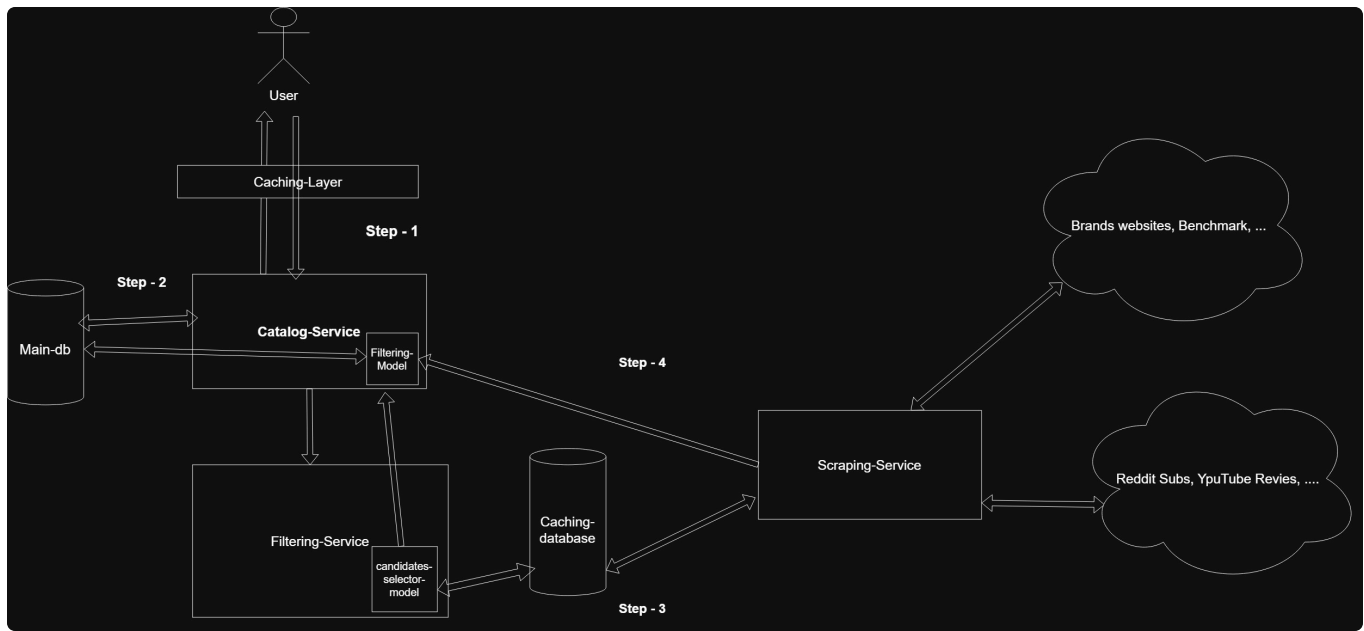




Step 1: Request Submission & Ingestion

The user submits a recommendation request containing their budget constraints, primary intended use, and qualitative feature priorities. The request passes through a **Caching Layer** and is ingested by the **Catalog Service**, initiating Phase 1.

User Request Payload:

JSON

```
{
  "maximum_budget_egp": 50000,
  "primary_use": "Programming & Software Development",
  "preferences": {
    "durability": "Important",
    "portability": "Nice to have",
    "battery_life": "Important",
    "performance": "Very important",
    "display_quality": "Nice to have",
    "quiet_operation": "Important",
    "keyboard": "Important",
    "connectivity_ports": "Important",
    "upgradeability": "Very important",
    "thermal_performance": "Very important"
  }
}
```

Step 2: Candidate Search & Preference Normalization

The **Catalog Service** translates the user's qualitative preferences into a normalized score scale (1–10). It queries the **Main Database** (`Main-db`) using these normalized scores and the defined price range to extract up to 15 initial candidate laptops matching the basic criteria. The candidate list is then passed to the **Filtering Service**.

Normalized Internal Payload:

JSON

```
{
  "maximum_budget_egg": 50000,
  "primary_use": "Programming & Software Development",
  "preference_scores": {
    "durability": 7,
    "portability": 4,
    "battery_life": 7,
    "performance": 10,
    "display_quality": 4,
    "quiet_operation": 7,
    "keyboard": 7,
    "connectivity_ports": 7,
    "upgradeability": 10,
    "thermal_performance": 10
  }
}
```

Step 3: Contextual Scraping & AI-Driven Selection (Phase 2)

The **Filtering Service** triggers the **Scraping Service** to gather real-time public sentiment and qualitative feedback (e.g., Reddit subreddits and YouTube review comments) for each of the 15 candidate laptops.

This aggregated data is evaluated by the **Candidates Selector Model** (AI Filtering Model). The model scores the contextual feedback against the user's specific priorities to select the top 3 optimal laptops, which are then routed back through the **Catalog Service** and **Caching Layer** to serve the final response to the user.

Step 4: Catalog Maintenance & Ingestion Jobs

- **Laptop Ingestion Pipeline:** A recurring background job in the **Catalog Service** periodically signals the **Scraping Service** (e.g., every 6 months) to crawl external brand websites, benchmark sources, and review platforms for newly released models. The scraped data is

refined and structured by the AI Filtering Model before being upserted into the **Main Database** (`Main-db`).

- **Price Sync Pipeline:** A separate, shorter-interval recurring job updates product pricing in the **Main Database** to ensure current market accuracy.